

SECTION 33 05 13 - MANHOLES AND STRUCTURES

PART 1 GENERAL

1.01 REFERENCE STANDARDS

- A. ASTM A48/A48M - Standard Specification for Gray Iron Castings 2003 (Reapproved 2016).
- B. ASTM A123/A123M - Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products 2017.
- C. ASTM C478 - Standard Specification for Circular Precast Reinforced Concrete Manhole Sections 2018.
- D. ASTM C478M - Standard Specification for Circular Precast Reinforced Concrete Manhole Sections (Metric) 2018.
- E. ASTM C923 - Standard Specification for Resilient Connectors Between Reinforced Concrete Manhole Structures, Pipes, and Laterals 2018.
- F. ASTM C923M - Standard Specification for Resilient Connectors Between Reinforced Concrete Manhole Structures, Pipes, and Laterals (Metric) 2018.

1.02 SUBMITTALS

- A. Product Data: Provide manhole covers, component construction, [_____] , features, configuration, and dimensions.
- B. Shop Drawings: Indicate manhole locations, elevations, piping sizes and elevations of penetrations.
- C. Manufacturer's Qualification Statement.

PART 2 PRODUCTS

2.01 MATERIALS

- A. Manhole Sections: Reinforced precast concrete in accordance with ASTM C478 (ASTM C478M), with resilient connectors complying with ASTM C923 (ASTM C923M).

2.02 COMPONENTS

- A. Lid and Frame: ASTM A48/A48M Class 30B Cast iron construction, machined flat bearing surface, removable lockable lid, closed lid design; live load rating of [_____] psf

([] kPa); sealing gasket; lid molded with identifying name. Provide []
manufactured by [] .

- B. Manhole Steps: Formed galvanized steel rungs; 3/4 inch (19 mm) diameter. Formed integral with manhole sections.
- C. Strap Anchors: Bent steel shape, [] x [] inch ([] x [] mm) size x [] inch ([] mm) thick, galvanized to ASTM A123/A123M Grade specified for applicable material category.

2.03 CONFIGURATION

END OF SECTION 33 05 13